Online Resource 1

Population and Environment. Author details are withheld for anonymous review.

Same country, different exits: place, settlement and the correlates of reported displacement in deltaic and floodplain Bangladesh

All tables use the analytic sample of 571 adults with a yes or no answer to item 5.13 unless stated otherwise. Interviewers are anonymised as E1 to E7.

Table S1. Answers to item 5.13 ('Have you been displaced from another area?') in the raw export, and their coding in the earlier cleaned file and in this analysis.

Answer	Barishal	Kurigram	Earlier cleaned file	This analysis
Yes	139	222	Displaced	Displaced
No	152	60	Not displaced	Not displaced
Not sure	2	0	Displaced	Excluded
No answer (item non-response)	9	18	Not displaced	Excluded
Aged under 18 (any answer)	1	2	Kept (merged into 18-35)	Excluded

Note. Raw KoboToolbox export: 603 interviews (303 Barishal, 300 Kurigram). One respondent reporting a gender other than woman or man was not in the earlier cleaned file (n = 602) and is not analysed. Rows are not mutually exclusive: of the three respondents under 18, one gave no answer to item 5.13. The earlier cleaned file was linked to the raw export record by record (602 of 602 linked; 563 agreed on all 35 shared fields and 39 on all but one). Results under the earlier coding and under both extreme codings of the 26 adult non-responses are in Table S6.

Table S2. Field sites: interviews, valid answers and reported displacement.

Site	Upazila	Interviews	Valid answers	Displaced, *n* (%; 95% CI)	Interviewers	Women, %	Resident >20 y, %	Dates
Laharhat (Tungibaria)	Barishal Sadar	40	38	1 (2.6; 0.5–13.5)	2	58	92	18 Sep–19 Sep 2023
Manta boat community, Laharhat	Barishal Sadar	58	58	17 (29.3; 19.2–42.0)	4	60	48	18 Sep–22 Sep 2023
Palashpur	Barishal Sadar	39	37	20 (54.1; 38.4–69.0)	6	28	67	16 Sep–21 Sep 2023
Bangabandhu Colony	Barishal Sadar	28	28	16 (57.1; 39.1–73.5)	6	61	68	17 Sep–17 Sep 2023
Namar Char	Barishal Sadar	37	34	12 (35.3; 21.5–52.1)	6	35	73	20 Sep–21 Sep 2023
Lalganj	Mehendiganj	42	38	26 (68.4; 52.5–80.9)	6	55	60	21 Sep–22 Sep 2023
Ulania (incl. Hasanpur)	Mehendiganj	58	57	46 (80.7; 68.7–88.9)	6	47	41	19 Sep–23 Sep 2023
Barishal, all sites		302	290	138 (47.6; 41.9–53.3)	6	49	62	16 Sep–23 Sep 2023
Modoner Char	Char Rajibpur	100	97	94 (96.9; 91.3–98.9)	5	50	26	18 Oct–22 Oct 2023
Baghmara	Rowmari	100	85	54 (63.5; 52.9–73.0)	5	50	63	16 Oct–17 Oct 2023
Faluar Char	Rowmari	100	99	74 (74.7; 65.4–82.3)	5	49	73	17 Oct–22 Oct 2023
Kurigram, all sites		300	281	222 (79.0; 73.9–83.4)	5	50	54	16 Oct–22 Oct 2023

Note. Sites assigned from the village field and the interview code. Wilson 95% confidence intervals. Women and residence percentages refer to all interviews at the site. Barishal Sadar sites: Laharhat (Tungibaria union), the Manta boat community at Laharhat, Palashpur, Bangabandhu Colony and Namar Char.

Table S3. Interviewers: allocation, respondents and reported displacement.

District	Interviewer	Valid answers	Displaced, *n* (%)	Sites	Women, %	Displaced / valid by site
Barishal	E1	54	6 (11.1)	6	17	BB 0/5; LG 0/7; LH 0/21; NC 0/5; PP 2/6; UL 4/10
Barishal	E2	47	35 (74.5)	6	83	BB 4/5; LG 7/7; MC 12/13; NC 2/7; PP 1/6; UL 9/9
Barishal	E3	52	25 (48.1)	6	10	BB 1/5; LG 6/7; MC 4/18; NC 5/6; PP 2/6; UL 7/10
Barishal	E4	42	21 (50.0)	6	83	BB 3/5; LG 5/6; MC 1/10; NC 0/6; PP 3/5; UL 9/10
Barishal	E5	47	28 (59.6)	6	38	BB 5/5; LG 3/5; LH 1/17; NC 4/4; PP 8/8; UL 7/8
Barishal	E6	48	23 (47.9)	6	79	BB 3/3; LG 5/6; MC 0/17; NC 1/6; PP 4/6; UL 10/10
Kurigram	E1	59	58 (98.3)	3	49	BG 18/19; FC 20/20; MD 20/20
Kurigram	E2	55	20 (36.4)	3	53	BG 0/15; FC 0/20; MD 20/20
Kurigram	E5	53	50 (94.3)	3	45	BG 11/14; FC 20/20; MD 19/19
Kurigram	E6	57	42 (73.7)	3	46	BG 12/19; FC 14/19; MD 16/19
Kurigram	E7	57	52 (91.2)	3	49	BG 13/18; FC 20/20; MD 19/19

Note. Site codes: LH Laharhat, MC Manta community, PP Palashpur, BB Bangabandhu Colony, NC Namar Char, LG Lalganj, UL Ulania (Barishal); BG Baghmara, FC Faluar Char, MD Modoner Char (Kurigram). Four interviewers worked in both districts. In Kurigram each interviewer was allocated 20 interviews per site, half with women.

Table S4. Item content of the exposure, hazard and non-economic loss counts, and prevalence by district.

Domain	Item	Barishal, *n* (%)	Kurigram, *n* (%)	Cramér's *V*	*p*
Exposure (location)	Flood-prone location	223 (76.9)	251 (89.3)	0.17	<0.001
Exposure (location)	Low-lying land	88 (30.3)	109 (38.8)	0.09	0.034
Exposure (location)	Erosion-prone location	95 (32.8)	164 (58.4)	0.26	<0.001
Exposure (location)	Salinity-affected location	13 (4.5)	1 (0.4)	0.13	0.001
Exposure (location)	Drought-prone location	7 (2.4)	215 (76.5)	0.76	<0.001
Rapid-onset hazard	Flood	211 (72.8)	276 (98.2)	0.36	<0.001
Rapid-onset hazard	Cyclone	153 (52.8)	29 (10.3)	0.46	<0.001
Rapid-onset hazard	Drought	5 (1.7)	221 (78.6)	0.79	<0.001
Rapid-onset hazard	Tidal surge	20 (6.9)	0 (0.0)	0.19	<0.001
Rapid-onset hazard	Heatwave	14 (4.8)	47 (16.7)	0.19	<0.001
Slow-onset hazard	Sea-level rise	57 (19.7)	36 (12.8)	0.09	0.027
Slow-onset hazard	Riverbank erosion	145 (50.0)	247 (87.9)	0.41	<0.001
Slow-onset hazard	Salinity intrusion	17 (5.9)	2 (0.7)	0.14	<0.001
Slow-onset hazard	Rising temperature	150 (51.7)	251 (89.3)	0.41	<0.001
Non-economic loss	Loss of cultural heritage	38 (13.1)	57 (20.3)	0.10	0.021
Non-economic loss	Adverse health impacts	221 (76.2)	213 (75.8)	0.00	0.910
Non-economic loss	Loss of biodiversity	69 (23.8)	83 (29.5)	0.06	0.121
Non-economic loss	Loss of social cohesion	44 (15.2)	25 (8.9)	0.10	0.021

Note. Barishal $n = 290$, Kurigram $n = 281$; p from Pearson χ^2 or Fisher tests. Exposure items are the reasons given for considering the current area vulnerable (items 2.4–2.5); hazard items are events respondents said they face in their area (3.2, 4.1); NELD items refer to the respondent or someone the respondent knows (5.2).

Table S5. Cell counts for categorical variables by district and reported displacement.

District	Variable	Level	Not displaced	Displaced
Barishal	Sex	Male	83	63
Barishal	Sex	Female	69	75
Barishal	Age group	18-35	52	56
Barishal	Age group	36-65	81	60
Barishal	Age group	65 or older	19	22
Barishal	Education	No education	79	49
Barishal	Education	Below primary	52	57
Barishal	Education	Secondary or above	21	32
Barishal	Rapid-onset impact	None or minor	36	25
Barishal	Rapid-onset impact	Recoverable	97	69
Barishal	Rapid-onset impact	Irrecoverable	19	44
Barishal	Slow-onset impact	None or minor	46	29
Barishal	Slow-onset impact	Recoverable	69	38
Barishal	Slow-onset impact	Irrecoverable	37	71
Barishal	Women's participation	No	19	53
Barishal	Women's participation	Not sure	9	9
Barishal	Women's participation	Yes	124	76
Barishal	Residence	Up to 5 years	1	10
Barishal	Residence	6-20 years	36	66
Barishal	Residence	Above 20 years	115	62
Barishal	Support received	In-kind relief	107	81
Barishal	Support received	Microcredit	16	5
Barishal	Support received	Training or other	11	3
Barishal	Support received	nan	18	49
Barishal	occ2	Manual labour	99	50
Barishal	occ2	Other	53	88
Kurigram	Sex	Male	28	117
Kurigram	Sex	Female	31	105
Kurigram	Age group	18-35	16	69
Kurigram	Age group	36-65	43	129
Kurigram	Age group	65 or older	0	24
Kurigram	Education	No education	29	75
Kurigram	Education	Below primary	16	87
Kurigram	Education	Secondary or above	14	60
Kurigram	Rapid-onset impact	None or minor	9	31
Kurigram	Rapid-onset impact	Recoverable	14	161
Kurigram	Rapid-onset impact	Irrecoverable	36	30
Kurigram	Slow-onset impact	None or minor	7	20
Kurigram	Slow-onset impact	Recoverable	9	127
Kurigram	Slow-onset impact	Irrecoverable	43	75
Kurigram	Women's participation	No	7	16
Kurigram	Women's participation	Not sure	1	39
Kurigram	Women's participation	Yes	51	167
Kurigram	Residence	Up to 5 years	2	22
Kurigram	Residence	6-20 years	14	94
Kurigram	Residence	Above 20 years	43	106
Kurigram	Support received	In-kind relief	19	64

District	Variable	Level	Not displaced	Displaced
Kurigram	Support received	Microcredit	3	47
Kurigram	Support received	Training or other	3	39
Kurigram	Support received	nan	34	72
Kurigram	occ2	Manual labour	26	100
Kurigram	occ2	Other	33	122

Note. No Kurigram respondent older than 65 years was non-displaced; estimates for that cell rest on the displaced respondents only.

Table S6. Pooled primary-model estimates across specifications and outcome codings (adjusted prevalence ratios, 95% CI).

Specification and term	aPR (95% CI)
Primary model (modified Poisson)	
Kurigram (ref. Barishal)	**1.48 (1.27–1.73)**
Rapid-onset hazard count	0.94 (0.83–1.07)
Slow-onset hazard count	**1.24 (1.13–1.35)**
Rapid-onset impact irrecoverable	0.88 (0.69–1.13)
Slow-onset impact irrecoverable	**1.35 (1.07–1.70)**
Strict outcome (work-motivated and unexplained moves excluded)	
Kurigram (ref. Barishal)	**1.77 (1.46–2.14)**
Rapid-onset hazard count	0.93 (0.80–1.08)
Slow-onset hazard count	**1.24 (1.13–1.37)**
Rapid-onset impact irrecoverable	0.94 (0.70–1.26)
Slow-onset impact irrecoverable	**1.52 (1.09–2.11)**
Alternative hazard taxonomy (sudden vs gradual)	
Kurigram (ref. Barishal)	**1.45 (1.23–1.71)**
Sudden-onset count (alternative)	**1.12 (1.02–1.22)**
Gradual count (alternative)	1.08 (0.99–1.18)
Rapid-onset impact irrecoverable	0.90 (0.70–1.15)
Slow-onset impact irrecoverable	**1.39 (1.10–1.75)**
Primary model + all correlates jointly	
Kurigram (ref. Barishal)	**1.38 (1.17–1.62)**
Rapid-onset hazard count	0.95 (0.85–1.07)
Slow-onset hazard count	**1.17 (1.06–1.29)**
Rapid-onset impact irrecoverable	0.98 (0.78–1.24)
Slow-onset impact irrecoverable	**1.40 (1.11–1.75)**
Original full model (35 terms)	
Kurigram (ref. Barishal)	**1.44 (1.22–1.71)**
Rapid-onset hazard count	0.92 (0.82–1.04)
Slow-onset hazard count	**1.16 (1.05–1.27)**
Earlier file coding (non-response = No, Not sure = Yes, minors kept; n = 602)	
Kurigram (ref. Barishal)	**1.44 (1.23–1.69)**
Rapid-onset hazard count	0.95 (0.83–1.08)
Slow-onset hazard count	**1.20 (1.09–1.31)**
Rapid-onset impact irrecoverable	0.95 (0.73–1.24)
Slow-onset impact irrecoverable	1.26 (0.99–1.62)
Bound: all item non-response coded not displaced	
Kurigram (ref. Barishal)	**1.47 (1.25–1.72)**
Rapid-onset hazard count	0.94 (0.83–1.07)
Slow-onset hazard count	**1.21 (1.11–1.33)**
Rapid-onset impact irrecoverable	0.97 (0.75–1.27)
Slow-onset impact irrecoverable	1.24 (0.97–1.58)
Bound: all item non-response coded displaced	
Kurigram (ref. Barishal)	**1.44 (1.25–1.66)**
Rapid-onset hazard count	0.95 (0.84–1.06)
Slow-onset hazard count	**1.23 (1.13–1.33)**
Rapid-onset impact irrecoverable	0.85 (0.67–1.07)
Slow-onset impact irrecoverable	**1.38 (1.10–1.72)**
Global district heterogeneity test	
Primary model (Primary model)	$\chi^2(11) = 111.7$, $p < 0.001$
Alternative hazard taxonomy	$\chi^2(11) = 115.4$, $p < 0.001$
Primary model, strict outcome	$\chi^2(11) = 113.7$, $p < 0.001$
Alternative grouping, by district	
Barishal: Sudden-onset count (alternative)	**1.41 (1.22–1.62)**
Barishal: Gradual count (alternative)	1.06 (0.91–1.25)
Kurigram: Sudden-onset count (alternative)	0.92 (0.83–1.02)
Kurigram: Gradual count (alternative)	0.99 (0.90–1.09)

Note. Stricter outcome: excludes displaced respondents who gave no reason for choosing their destination or chose it for work (n = 521). Earlier coding: n = 602, item non-response coded not displaced and 'not sure' coded displaced, minors retained. Bounds: the 26 adult non-responses coded all not displaced or all displaced (n = 597). Bold: 95% CI excludes 1.

Table S7. Correlates added singly to the district-specific primary model, without and with site and interviewer fixed effects.

District	Correlate	Primary model + correlate	+ site and interviewer fixed effects
Barishal	NELD count	**0.78 (0.65–0.94)**	0.83 (0.63–1.10)
Barishal	Women in response decisions: not sure (ref. no)	0.99 (0.61–1.60)	1.01 (0.72–1.43)

District	Correlate	Primary model + correlate	+ site and interviewer fixed effects
Barishal	Women in response decisions: yes (ref. no)	**0.62 (0.49–0.79)**	**0.70 (0.51–0.96)**
Barishal	Exposure of current location	1.04 (0.89–1.21)	1.00 (0.80–1.25)
Barishal	Residence 6–20 y (ref. ≤5)	0.94 (0.65–1.35)	0.92 (0.54–1.58)
Barishal	Residence >20 y (ref. ≤5)	**0.55 (0.34–0.87)**	0.63 (0.35–1.12)
Barishal	Microcredit (ref. in-kind relief)	0.56 (0.25–1.26)	0.75 (0.28–2.00)
Barishal	Training or other (ref. in-kind)	0.43 (0.14–1.29)	0.63 (0.19–2.07)
Barishal	No support (ref. in-kind relief)	**1.32 (1.01–1.71)**	1.03 (0.66–1.61)
Barishal	Non-manual occupation (ref. manual)	**1.42 (1.08–1.87)**	1.02 (0.68–1.51)
Kurigram	NELD count	0.96 (0.89–1.04)	0.97 (0.81–1.16)
Kurigram	Women in response decisions: not sure (ref. no)	**1.39 (1.07–1.79)**	1.26 (0.78–2.02)
Kurigram	Women in response decisions: yes (ref. no)	1.15 (0.88–1.49)	1.22 (0.72–2.06)
Kurigram	Exposure of current location	1.05 (0.98–1.13)	1.05 (0.96–1.14)
Kurigram	Residence 6–20 y (ref. ≤5)	1.04 (0.91–1.20)	1.00 (0.70–1.45)
Kurigram	Residence >20 y (ref. ≤5)	**0.77 (0.66–0.91)**	0.73 (0.44–1.19)
Kurigram	Microcredit (ref. in-kind relief)	1.06 (0.91–1.24)	0.87 (0.61–1.25)
Kurigram	Training or other (ref. in-kind)	1.07 (0.92–1.25)	0.98 (0.80–1.20)
Kurigram	No support (ref. in-kind relief)	0.96 (0.82–1.13)	0.91 (0.73–1.14)
Kurigram	Non-manual occupation (ref. manual)	0.95 (0.82–1.10)	0.85 (0.68–1.07)

Note. Adjusted prevalence ratios (95% CI). Fixed-effects intervals are CR2 cluster-robust for site-by-interviewer cells with Satterthwaite degrees of freedom. Bold: 95% CI excludes 1.

Table S8. Leave-one-interviewer-out analysis.

District	Term	Range of aPR across runs	Runs with 95% CI excluding 1
Barishal	Age 36–65 (ref. 18–35)	0.83 to 0.98	0 of 7
Barishal	Age >65 (ref. 18–35)	1.09 to 1.51	0 of 7
Barishal	Below primary (ref. none)	1.62 to 2.42	7 of 7
Barishal	Secondary or above (ref. none)	1.42 to 2.46	7 of 7
Barishal	Woman (ref. man)	1.17 to 2.17	2 of 7
Barishal	Rapid-onset impact irrecoverable	0.79 to 1.61	2 of 7
Barishal	Rapid-onset impact recoverable	0.77 to 1.40	0 of 7
Barishal	Slow-onset impact irrecoverable	1.62 to 2.44	7 of 7
Barishal	Slow-onset impact recoverable	0.82 to 1.35	0 of 7
Barishal	Rapid-onset hazard count	0.78 to 1.09	0 of 7
Barishal	Slow-onset hazard count	1.42 to 1.58	7 of 7
Kurigram	Age 36–65 (ref. 18–35)	1.03 to 1.17	0 of 7
Kurigram	Age >65 (ref. 18–35)	1.20 to 1.41	7 of 7
Kurigram	Below primary (ref. none)	1.06 to 1.14	0 of 7
Kurigram	Secondary or above (ref. none)	1.02 to 1.15	0 of 7
Kurigram	Woman (ref. man)	0.94 to 0.99	0 of 7
Kurigram	Rapid-onset impact irrecoverable	0.54 to 0.98	5 of 7
Kurigram	Rapid-onset impact recoverable	1.05 to 1.21	0 of 7
Kurigram	Slow-onset impact irrecoverable	0.91 to 1.92	0 of 7
Kurigram	Slow-onset impact recoverable	1.18 to 2.00	1 of 7
Kurigram	Rapid-onset hazard count	0.91 to 1.04	0 of 7
Kurigram	Slow-onset hazard count	0.85 to 0.95	1 of 7
Global test	Interviewer E1 left out	$\chi^2(11) = 87.2$	<0.001
Global test	Interviewer E2 left out	$\chi^2(11) = 44.7$	<0.001
Global test	Interviewer E3 left out	$\chi^2(11) = 124.3$	<0.001
Global test	Interviewer E4 left out	$\chi^2(11) = 121.8$	<0.001
Global test	Interviewer E5 left out	$\chi^2(11) = 111.8$	<0.001
Global test	Interviewer E6 left out	$\chi^2(11) = 90.7$	<0.001
Global test	Interviewer E7 left out	$\chi^2(11) = 91.8$	<0.001

Note. Each run drops one interviewer's respondents and refits the district-specific primary models and the pooled global heterogeneity test. Interviewers working in one district only affect that district's estimates.

Table S9. Dependence of each item on the interviewer (Cramér's *V*).

District	Item	*V*, interviewer	*V*, within site (weighted)	*p*
Barishal	Reported displacement (5.13)	0.39	0.62	<0.001
Barishal	NELD: displacement or migration (5.2)	0.50	0.68	<0.001
Barishal	Sex	0.63	0.77	<0.001
Barishal	Age group	0.27	0.44	<0.001
Barishal	Education	0.21	0.44	0.005
Barishal	Rapid-onset impact severity (3.3)	0.54	0.63	<0.001
Barishal	Slow-onset impact severity (4.2)	0.63	0.74	<0.001
Barishal	Rapid-onset hazard count (3.2)	0.28	0.61	<0.001
Barishal	Slow-onset hazard count (4.1)	0.28	0.45	<0.001
Barishal	Current-location exposure count (2.5)	0.27	0.51	<0.001
Barishal	NELD count (5.2)	0.35	0.54	<0.001
Barishal	Women in response decisions (5.9)	0.38	0.50	<0.001
Barishal	Residence duration (2.1)	0.41	0.54	<0.001
Barishal	Support received (7.1)	0.24	0.51	<0.001
Kurigram	Reported displacement (5.13)	0.56	0.64	<0.001
Kurigram	NELD: displacement or migration (5.2)	0.32	0.35	<0.001
Kurigram	Sex	0.05	0.08	0.937
Kurigram	Age group	0.29	0.37	<0.001
Kurigram	Education	0.35	0.41	<0.001

District	Item	*V*, interviewer	*V*, within site (weighted)	*p*
Kurigram	Rapid-onset impact severity (3.3)	0.80	0.80	<0.001
Kurigram	Slow-onset impact severity (4.2)	0.66	0.67	<0.001
Kurigram	Rapid-onset hazard count (3.2)	0.26	0.41	<0.001
Kurigram	Slow-onset hazard count (4.1)	0.45	0.53	<0.001
Kurigram	Current-location exposure count (2.5)	0.45	0.51	<0.001
Kurigram	NELD count (5.2)	0.36	0.43	<0.001
Kurigram	Women in response decisions (5.9)	0.42	0.46	<0.001
Kurigram	Residence duration (2.1)	0.37	0.49	<0.001
Kurigram	Support received (7.1)	0.43	0.48	<0.001

Note. Within-site values are averages over sites weighted by site size; with few respondents per interviewer and site they are biased upwards, so comparisons between items are more informative than absolute values. Respondent sex was allocated by interviewer in Barishal by design. *p*, Monte Carlo χ^2 test (2,000 replicates).

Table S10. Agreement between item 5.13 and the 'displacement or migration' option of item 5.2.

District	Interviewer	*n*	Both / 5.13 only / 5.2 only / neither; or % yes	Agreement, %	κ
Barishal	All	290	107/31/44/108	74.1	0.48
Kurigram	All	281	198/24/41/18	76.9	0.22
Barishal	E1	54	5.13: 11%; 5.2: 9%	90.7	
Barishal	E2	47	5.13: 74%; 5.2: 49%	74.5	
Barishal	E3	52	5.13: 48%; 5.2: 73%	59.6	
Barishal	E4	42	5.13: 50%; 5.2: 50%	71.4	
Barishal	E5	47	5.13: 60%; 5.2: 87%	59.6	
Barishal	E6	48	5.13: 48%; 5.2: 48%	87.5	
Kurigram	E1	59	5.13: 98%; 5.2: 98%	96.6	
Kurigram	E2	55	5.13: 36%; 5.2: 84%	41.8	
Kurigram	E5	53	5.13: 94%; 5.2: 98%	92.5	
Kurigram	E6	57	5.13: 74%; 5.2: 70%	86.0	
Kurigram	E7	57	5.13: 91%; 5.2: 75%	66.7	

Note. Item 5.2 asked whether the respondent or someone the respondent knows had experienced non-economic losses due to climate change, one option being displacement or migration.

Table S11. Residence at the current address among displaced respondents, a bound on time since arrival.

District	Site	Displaced, *n*	≤5 years, *n* (%)	6–20 years, *n* (%)	>20 years, *n* (%)
Barishal	All sites	138	10 (7)	66 (48)	62 (45)
Barishal	Bangabandhu Colony	16	3 (19)	4 (25)	9 (56)
Barishal	Laharhat (Tungibaria)	1	0 (0)	0 (0)	1 (100)
Barishal	Lalganj	26	5 (19)	9 (35)	12 (46)
Barishal	Manta boat community, Laharhat	17	1 (6)	12 (71)	4 (24)
Barishal	Namar Char	12	0 (0)	3 (25)	9 (75)
Barishal	Palashpur	20	1 (5)	8 (40)	11 (55)
Barishal	Ulania (incl. Hasanpur)	46	0 (0)	30 (65)	16 (35)
Kurigram	All sites	222	22 (10)	94 (42)	106 (48)
Kurigram	Baghmara	54	6 (11)	16 (30)	32 (59)
Kurigram	Faluar Char	74	4 (5)	20 (27)	50 (68)
Kurigram	Modoner Char	94	12 (13)	58 (62)	24 (26)

Note. Item 2.1, 'How long have you been living here?'. If respondents moved once, residence duration bounds the time since displacement; repeated moves would make the last move more recent.

Table S12. Discrimination, calibration and cross-validated comparison with flexible learners.

Model	AUC	AUC, optimism-corrected	Slope, corrected	Brier	Nagelkerke R^2, apparent / corrected
Primary model (Primary model)	0.757	0.734	0.88	0.188	0.261 / 0.215
Primary model with district interactions	0.846	0.817	0.83	0.151	0.457 / 0.387
Pooled: Logistic regression (CV, full variable set)	0.796			0.171	
Pooled: Random forest (CV, full variable set)	0.883			0.138	
Pooled: Gradient boosting (CV, full variable set)	0.858			0.141	
Barishal: Logistic regression (CV, full variable set)	0.794			0.197	
Barishal: Random forest (CV, full variable set)	0.845			0.168	
Barishal: Gradient boosting (CV, full variable set)	0.843			0.165	
Kurigram: Logistic regression (CV, full variable set)	0.803			0.128	
Kurigram: Random forest (CV, full variable set)	0.835			0.103	
Kurigram: Gradient boosting (CV, full variable set)	0.838			0.101	
Pooled: Logistic, additive (unpenalised) (CV)	0.796				
Pooled: Logistic, additive (ridge) (CV)	0.787				
Pooled: Logistic, district interactions (ridge) (CV)	0.850				
Pooled: Random forest (CV)	0.883				

Note. Optimism correction by 1,000 bootstrap resamples. Cross-validation: 10 repetitions of stratified 10-fold cross-validation (5-fold within districts) on the full variable set, with tuning inside training folds.

Table S13. Model-standardised probabilities of reported displacement (%), by district.

District	Variable	Level	Probability (95% CI)	Model
Barishal	Slow-onset count	0	23 (15–32)	Primary model

District	Variable	Level	Probability (95% CI)	Model
Barishal	Slow-onset count	1	41 (36–47)	Primary model
Barishal	Slow-onset count	2	62 (54–70)	Primary model
Barishal	Slow-onset count	3	79 (68–89)	Primary model
Barishal	Rapid-onset count	1	49 (43–55)	Primary model
Barishal	Rapid-onset count	2	45 (38–52)	Primary model
Barishal	Rapid-onset count	3	41 (27–54)	Primary model
Barishal	Rapid-onset impact	None or minor	45 (34–56)	Primary model
Barishal	Rapid-onset impact	Recoverable	44 (37–51)	Primary model
Barishal	Rapid-onset impact	Irrecoverable	60 (47–73)	Primary model
Barishal	Slow-onset impact	None or minor	36 (26–46)	Primary model
Barishal	Slow-onset impact	Recoverable	36 (28–45)	Primary model
Barishal	Slow-onset impact	Irrecoverable	66 (57–73)	Primary model
Barishal	NELD count	0	59 (49–69)	Primary model + correlate
Barishal	NELD count	1	50 (45–55)	Primary model + correlate
Barishal	NELD count	2	41 (33–49)	Primary model + correlate
Barishal	NELD count	3	32 (20–46)	Primary model + correlate
Barishal	Women's participation	No	65 (54–76)	Primary model + correlate
Barishal	Women's participation	Not sure	56 (34–74)	Primary model + correlate
Barishal	Women's participation	Yes	41 (35–47)	Primary model + correlate
Barishal	Exposure count	0	42 (29–53)	Primary model + correlate
Barishal	Exposure count	1	46 (39–51)	Primary model + correlate
Barishal	Exposure count	2	50 (43–56)	Primary model + correlate
Barishal	Exposure count	3	53 (40–67)	Primary model + correlate
Barishal	Exposure count	4	57 (37–77)	Primary model + correlate
Barishal	Residence	Up to 5 years	75 (45–95)	Primary model + correlate
Barishal	Residence	6-20 years	60 (51–69)	Primary model + correlate
Barishal	Residence	Above 20 years	38 (31–46)	Primary model + correlate
Barishal	Support received	In-kind relief	47 (40–53)	Primary model + correlate
Barishal	Support received	Microcredit	30 (11–52)	Primary model + correlate
Barishal	Support received	Training or other	21 (4–43)	Primary model + correlate
Barishal	Support received	None	63 (51–74)	Primary model + correlate
Kurigram	Slow-onset count	0	88 (72–97)	Primary model
Kurigram	Slow-onset count	1	83 (75–91)	Primary model
Kurigram	Slow-onset count	2	77 (73–81)	Primary model
Kurigram	Slow-onset count	3	70 (57–82)	Primary model
Kurigram	Rapid-onset count	1	77 (66–89)	Primary model
Kurigram	Rapid-onset count	2	78 (74–82)	Primary model
Kurigram	Rapid-onset count	3	79 (67–88)	Primary model
Kurigram	Rapid-onset impact	None or minor	75 (58–91)	Primary model
Kurigram	Rapid-onset impact	Recoverable	90 (84–94)	Primary model
Kurigram	Rapid-onset impact	Irrecoverable	56 (42–69)	Primary model
Kurigram	Slow-onset impact	None or minor	66 (45–86)	Primary model
Kurigram	Slow-onset impact	Recoverable	87 (78–94)	Primary model
Kurigram	Slow-onset impact	Irrecoverable	74 (66–82)	Primary model
Kurigram	NELD count	0	81 (71–89)	Primary model + correlate
Kurigram	NELD count	1	79 (74–83)	Primary model + correlate
Kurigram	NELD count	2	76 (71–82)	Primary model + correlate
Kurigram	NELD count	3	74 (61–85)	Primary model + correlate
Kurigram	Women's participation	No	67 (47–84)	Primary model + correlate
Kurigram	Women's participation	Not sure	93 (84–98)	Primary model + correlate
Kurigram	Women's participation	Yes	77 (72–82)	Primary model + correlate
Kurigram	Exposure count	0	50 (30–73)	Primary model + correlate
Kurigram	Exposure count	1	62 (49–75)	Primary model + correlate
Kurigram	Exposure count	2	73 (67–78)	Primary model + correlate
Kurigram	Exposure count	3	81 (77–85)	Primary model + correlate
Kurigram	Exposure count	4	88 (81–94)	Primary model + correlate
Kurigram	Residence	Up to 5 years	88 (75–98)	Primary model + correlate
Kurigram	Residence	6-20 years	87 (81–93)	Primary model + correlate
Kurigram	Residence	Above 20 years	68 (62–74)	Primary model + correlate
Kurigram	Support received	In-kind relief	76 (67–84)	Primary model + correlate
Kurigram	Support received	Microcredit	84 (73–96)	Primary model + correlate
Kurigram	Support received	Training or other	85 (74–96)	Primary model + correlate
Kurigram	Support received	None	76 (69–82)	Primary model + correlate

Note. g-computation from Firth-penalised district models; percentile intervals from 1,000 bootstrap resamples within district.

Table S14. E-values for the pooled primary-model prevalence ratios.

Term	aPR	E-value, estimate	E-value, CI limit
Woman (ref. man)	1.18	1.63	1.27
Age 36–65 (ref. 18–35)	0.91	1.45	1.00
Age >65 (ref. 18–35)	1.27	1.85	1.28
Below primary (ref. none)	1.28	1.87	1.44
Secondary or above (ref. none)	1.30	1.93	1.43
Kurigram (ref. Barishal)	1.48	2.33	1.86
Rapid-onset hazard count	0.94	1.32	1.00
Slow-onset hazard count	1.24	1.77	1.52

Term	aPR	E-value, estimate	E-value, CI limit
Rapid-onset impact recoverable	1.04	1.25	1.00
Rapid-onset impact irrecoverable	0.88	1.52	1.00
Slow-onset impact recoverable	1.20	1.69	1.00
Slow-onset impact irrecoverable	1.35	2.03	1.34

Note. E-value of 1 for the CI limit: the interval already includes the null.

Table S15. Primary model as odds ratios: logistic and Firth-penalised logistic regression.

Term	Pooled, logistic OR	Pooled, Firth OR	Barishal, Firth OR	Kurigram, Firth OR
Woman (ref. man)	1.77 (1.17–2.70)	1.74 (1.16–2.64)	2.22 (1.18–4.26)	0.88 (0.44–1.74)
Age 36–65 (ref. 18–35)	0.67 (0.42–1.06)	0.68 (0.43–1.06)	0.70 (0.37–1.32)	1.75 (0.77–4.02)
Age >65 (ref. 18–35)	2.02 (1.00–4.20)	1.97 (0.98–4.03)	1.58 (0.65–3.84)	22.19 (2.27–3021.20)
Below primary (ref. none)	2.03 (1.29–3.19)	1.99 (1.28–3.12)	3.36 (1.75–6.67)	1.73 (0.77–3.92)
Secondary or above (ref. none)	2.27 (1.30–4.03)	2.21 (1.27–3.89)	3.93 (1.69–9.54)	1.52 (0.61–3.83)
Kurigram (ref. Barishal)	3.58 (2.26–5.73)	3.47 (2.21–5.51)		
Rapid-onset hazard count	0.75 (0.52–1.07)	0.76 (0.53–1.07)	0.79 (0.49–1.24)	1.08 (0.53–2.23)
Slow-onset hazard count	1.94 (1.45–2.64)	1.91 (1.43–2.57)	2.84 (1.90–4.42)	0.61 (0.28–1.27)
Rapid-onset impact recoverable	1.23 (0.70–2.15)	1.23 (0.70–2.13)	0.92 (0.42–2.00)	3.16 (1.05–9.34)
Rapid-onset impact irrecoverable	0.62 (0.31–1.25)	0.64 (0.32–1.26)	2.15 (0.85–5.51)	0.39 (0.11–1.26)
Slow-onset impact recoverable	1.74 (0.95–3.19)	1.71 (0.94–3.11)	1.02 (0.47–2.22)	4.27 (1.21–14.91)
Slow-onset impact irrecoverable	2.36 (1.24–4.56)	2.30 (1.22–4.38)	4.73 (2.06–11.28)	1.63 (0.46–5.64)

Note. Profile-likelihood 95% CI. With a common outcome odds ratios exceed the corresponding prevalence ratios.

Table S16. Exploratory item-level models.

Set	Sample	Item	Reporting, *n*	aPR (95% CI)
Hazards	Pooled	Flood	487	**2.36 (1.46–3.82)**
Hazards	Pooled	Cyclone	182	**0.79 (0.66–0.96)**
Hazards	Pooled	Drought	226	1.13 (0.91–1.40)
Hazards	Pooled	Heatwave	61	**0.72 (0.54–0.96)**
Hazards	Pooled	Sea-level rise	93	1.00 (0.87–1.15)
Hazards	Pooled	Riverbank erosion	392	**1.47 (1.21–1.78)**
Hazards	Pooled	Rising temperature	401	0.99 (0.84–1.17)
Hazards	Barishal	Flood	211	**1.93 (1.06–3.54)**
Hazards	Barishal	Cyclone	153	**0.73 (0.54–0.97)**
Hazards	Barishal	Heatwave	14	0.82 (0.39–1.71)
Hazards	Barishal	Sea-level rise	57	1.06 (0.77–1.44)
Hazards	Barishal	Riverbank erosion	145	**2.30 (1.66–3.20)**
Hazards	Barishal	Rising temperature	150	1.26 (1.00–1.58)
Hazards	Kurigram	Flood	276	1.08 (0.72–1.60)
Hazards	Kurigram	Cyclone	29	0.91 (0.74–1.12)
Hazards	Kurigram	Drought	221	1.11 (0.91–1.35)
Hazards	Kurigram	Heatwave	47	0.99 (0.76–1.28)
Hazards	Kurigram	Sea-level rise	36	1.01 (0.92–1.10)
Hazards	Kurigram	Riverbank erosion	247	**0.88 (0.79–0.98)**
Hazards	Kurigram	Rising temperature	251	0.87 (0.67–1.13)
Exposure	Pooled	Flood-prone location	474	**0.84 (0.70–1.00)**
Exposure	Pooled	Low-lying land	197	1.13 (1.00–1.28)
Exposure	Pooled	Erosion-prone location	259	1.04 (0.92–1.19)
Exposure	Pooled	Drought-prone location	222	1.12 (0.95–1.32)
Exposure	Barishal	Flood-prone location	223	1.09 (0.83–1.43)
Exposure	Barishal	Low-lying land	88	0.95 (0.75–1.20)
Exposure	Barishal	Erosion-prone location	95	1.07 (0.81–1.41)
Exposure	Kurigram	Flood-prone location	251	**0.78 (0.68–0.89)**
Exposure	Kurigram	Low-lying land	109	**1.13 (1.00–1.27)**
Exposure	Kurigram	Erosion-prone location	164	1.07 (0.96–1.20)
Exposure	Kurigram	Drought-prone location	215	1.16 (1.00–1.34)
NELD	Pooled	Loss of cultural heritage	95	0.93 (0.78–1.11)
NELD	Pooled	Adverse health impacts	434	**0.83 (0.71–0.97)**
NELD	Pooled	Loss of biodiversity	152	0.88 (0.75–1.02)
NELD	Pooled	Loss of social cohesion	69	0.81 (0.65–1.02)
NELD	Barishal	Loss of cultural heritage	38	0.92 (0.62–1.37)
NELD	Barishal	Adverse health impacts	221	1.07 (0.79–1.43)
NELD	Barishal	Loss of biodiversity	69	**0.54 (0.36–0.81)**
NELD	Barishal	Loss of social cohesion	44	0.73 (0.48–1.10)
NELD	Kurigram	Loss of cultural heritage	57	**0.82 (0.68–0.98)**
NELD	Kurigram	Adverse health impacts	213	0.94 (0.80–1.11)
NELD	Kurigram	Loss of biodiversity	83	1.02 (0.89–1.16)
NELD	Kurigram	Loss of social cohesion	25	1.14 (0.87–1.48)

Note. Each set enters its items in place of (hazards) or in addition to (exposure, NELD) the corresponding count, with sex, age, education and impact severity. Items reported by fewer than ten respondents in a sample are omitted.

Table S17. Internal consistency of the counts (Kuder-Richardson 20) and collinearity of the primary model.

Count or variable	Pooled / GVIF	Barishal / df	Kurigram / scaled
Exposure (5 items)	0.26	0.03	0.14
Rapid-onset hazards (5)	-0.69	-0.24	-1.08
Slow-onset hazards (4)	0.10	0.07	0.03
NELD (4)	-0.22	-0.12	-0.32
GVIF: Sex	1.20	df 1	scaled 1.10

Count or variable	Pooled / GVIF	Barishal / df	Kurigram / scaled
GVIF: Age group	1.30	df 2	scaled 1.07
GVIF: Education	1.32	df 2	scaled 1.07
GVIF: Rapid-onset impact	1.85	df 2	scaled 1.17
GVIF: Slow-onset impact	1.94	df 2	scaled 1.18
GVIF: district	1.43	df 1	scaled 1.20
GVIF: Rapid-onset count	1.70	df 1	scaled 1.30
GVIF: Slow-onset count	1.48	df 1	scaled 1.22

Note. Low or negative KR-20 values show that the items do not form reflective scales, which they were not designed to be; they are not a test of the counts' validity. GVIF scaled as $GVIF^{1/(2df)}$.

Table S18. Transportability of Firth-penalised models across places.

Predictors	Model	*n*	Observed, %	Predicted, %	O/E	Cal. intercept	Cal. slope	AUC (95% CI)
Primary model	Barishal -> Barishal (10-fold CV)	290	47.6		1.00			0.76 (0.75–0.77)
Primary model	Kurigram -> Kurigram (10-fold CV)	281	79.0		1.01			0.78 (0.76–0.79)
Primary model	Barishal -> Kurigram	281	79.0	59.5	1.33	1.22	-0.61	0.31 (0.23–0.38)
Primary model	Kurigram -> Barishal	290	47.6	78.2	0.61	-2.01	-0.23	0.39 (0.33–0.46)
Primary model	Pooled (both districts) -> Sunamganj 2024	128	18.0	68.2	0.26	-2.97	0.32	0.64 (0.52–0.76)
Primary model	Barishal -> Sunamganj 2024	128	18.0	43.2	0.42	-2.12	0.22	0.63 (0.51–0.75)
Primary model	Kurigram -> Sunamganj 2024	128	18.0	88.5	0.20	-4.54	-0.02	0.45 (0.31–0.59)
Sex, age, education, hazard counts	Barishal -> Barishal (10-fold CV)	290	47.6		1.00			0.69 (0.68–0.70)
Sex, age, education, hazard counts	Kurigram -> Kurigram (10-fold CV)	281	79.0		1.01			0.55 (0.53–0.57)
Sex, age, education, hazard counts	Barishal -> Kurigram	281	79.0	54.6	1.45	1.27	0.24	0.55 (0.48–0.62)
Sex, age, education, hazard counts	Kurigram -> Barishal	290	47.6	80.6	0.59	-1.69	0.17	0.55 (0.49–0.62)
Sex, age, education, hazard counts	Pooled (both districts) -> Sunamganj 2024	128	18.0	65.8	0.27	-2.66	0.47	0.66 (0.55–0.77)

Note. Within-district rows: 20 repetitions of 10-fold cross-validation (mean, 2.5th–97.5th percentiles). Transported rows: DeLong intervals. A calibration slope near or below zero means the ranking does not transport.

Table S19. District-specific primary-model estimates under design-based specifications (values for Fig. 6).

District	Term	Primary (HC0)	CR2 clusters	+ site FE	+ interviewer FE	+ both (CR2)
Barishal	Woman (ref. man)	**1.42 (1.09–1.84)**	1.42 (0.91–2.20)	**1.55 (1.20–2.00)**	1.04 (0.78–1.38)	1.39 (0.99–1.96)
Barishal	Age 36–65 (ref. 18–35)	0.91 (0.69–1.20)	0.91 (0.64–1.29)	0.86 (0.67–1.09)	1.01 (0.78–1.30)	0.94 (0.72–1.22)
Barishal	Age >65 (ref. 18–35)	1.33 (0.93–1.90)	1.33 (0.83–2.12)	1.05 (0.78–1.42)	1.29 (0.92–1.81)	1.03 (0.70–1.51)
Barishal	Below primary (ref. none)	**1.84 (1.41–2.39)**	**1.84 (1.10–3.08)**	1.19 (0.92–1.56)	**1.72 (1.33–2.23)**	1.20 (0.84–1.73)
Barishal	Secondary or above (ref. none)	**1.75 (1.30–2.36)**	1.75 (0.95–3.23)	1.14 (0.84–1.54)	**1.58 (1.19–2.09)**	1.12 (0.73–1.73)
Barishal	Rapid-onset hazard count	0.93 (0.74–1.17)	0.93 (0.58–1.51)	0.92 (0.73–1.16)	0.98 (0.77–1.24)	1.04 (0.61–1.77)
Barishal	Slow-onset hazard count	**1.53 (1.33–1.77)**	**1.53 (1.16–2.03)**	**1.30 (1.10–1.53)**	**1.59 (1.39–1.82)**	**1.38 (1.08–1.76)**
Barishal	Rapid-onset impact recoverable	0.96 (0.67–1.36)	0.96 (0.53–1.74)	1.08 (0.77–1.51)	1.22 (0.88–1.69)	1.29 (0.72–2.33)
Barishal	Rapid-onset impact irrecoverable	1.33 (0.91–1.95)	1.33 (0.66–2.68)	1.22 (0.87–1.71)	1.48 (0.96–2.28)	1.53 (0.70–3.37)
Barishal	Slow-onset impact recoverable	1.00 (0.69–1.45)	1.00 (0.61–1.64)	1.09 (0.77–1.54)	1.49 (0.99–2.25)	1.41 (0.93–2.15)
Barishal	Slow-onset impact irrecoverable	**1.97 (1.36–2.87)**	**1.97 (1.17–3.32)**	**2.05 (1.41–2.98)**	**2.03 (1.36–3.02)**	**1.74 (1.01–2.97)**
Kurigram	Woman (ref. man)	0.97 (0.88–1.08)	0.97 (0.89–1.07)	0.97 (0.88–1.08)	0.98 (0.88–1.08)	0.97 (0.89–1.05)
Kurigram	Age 36–65 (ref. 18–35)	1.09 (0.95–1.26)	1.09 (0.93–1.28)	1.09 (0.95–1.25)	1.14 (0.99–1.31)	1.13 (0.96–1.33)
Kurigram	Age >65 (ref. 18–35)	**1.33 (1.13–1.57)**	**1.33 (1.03–1.72)**	**1.30 (1.09–1.55)**	**1.23 (1.03–1.46)**	1.17 (0.79–1.75)
Kurigram	Below primary (ref. none)	1.11 (0.96–1.28)	1.11 (0.87–1.41)	1.08 (0.94–1.25)	1.03 (0.90–1.18)	1.00 (0.82–1.22)
Kurigram	Secondary or above (ref. none)	1.11 (0.93–1.32)	1.11 (0.87–1.42)	1.07 (0.90–1.27)	1.06 (0.88–1.29)	1.01 (0.73–1.40)
Kurigram	Rapid-onset hazard count	1.01 (0.90–1.14)	1.01 (0.88–1.17)	1.06 (0.94–1.19)	1.01 (0.90–1.13)	1.05 (0.88–1.25)
Kurigram	Slow-onset hazard count	0.92 (0.84–1.00)	0.92 (0.82–1.04)	0.92 (0.84–1.01)	0.91 (0.82–1.01)	0.91 (0.76–1.10)
Kurigram	Rapid-onset impact recoverable	1.20 (0.99–1.44)	1.20 (0.78–1.83)	1.20 (0.99–1.45)	0.96 (0.72–1.28)	1.00 (0.65–1.55)
Kurigram	Rapid-onset impact irrecoverable	**0.63 (0.45–0.88)**	0.63 (0.16–2.45)	**0.64 (0.47–0.87)**	0.90 (0.59–1.37)	1.00 (0.58–1.72)
Kurigram	Slow-onset impact recoverable	1.25 (0.99–1.59)	1.25 (0.94–1.68)	1.24 (0.97–1.58)	1.09 (0.81–1.48)	1.10 (0.75–1.64)
Kurigram	Slow-onset impact irrecoverable	1.13 (0.87–1.46)	1.13 (0.81–1.58)	1.09 (0.85–1.40)	0.98 (0.71–1.34)	0.97 (0.64–1.45)

Note. Adjusted prevalence ratios (95% CI). CR2 clusters are site-by-interviewer cells (Barishal 36, Kurigram 15), with Satterthwaite degrees of freedom. Bold: 95% CI excludes 1.

Table S20. Sunamganj round (31 October to 10 November 2024): answers to item 5.13 by upazila.

Upazila	Yes	No	No answer	Displaced among valid answers, %
Bishwambarpur	5	49	39	9.3
Tahirpur	18	56	21	24.3

Note. Same form version as the 2023 rounds. 188 interviews; 128 adults with a valid answer were used in the transportability analysis. The high item non-response (32%) and the one-year gap limit the comparison.

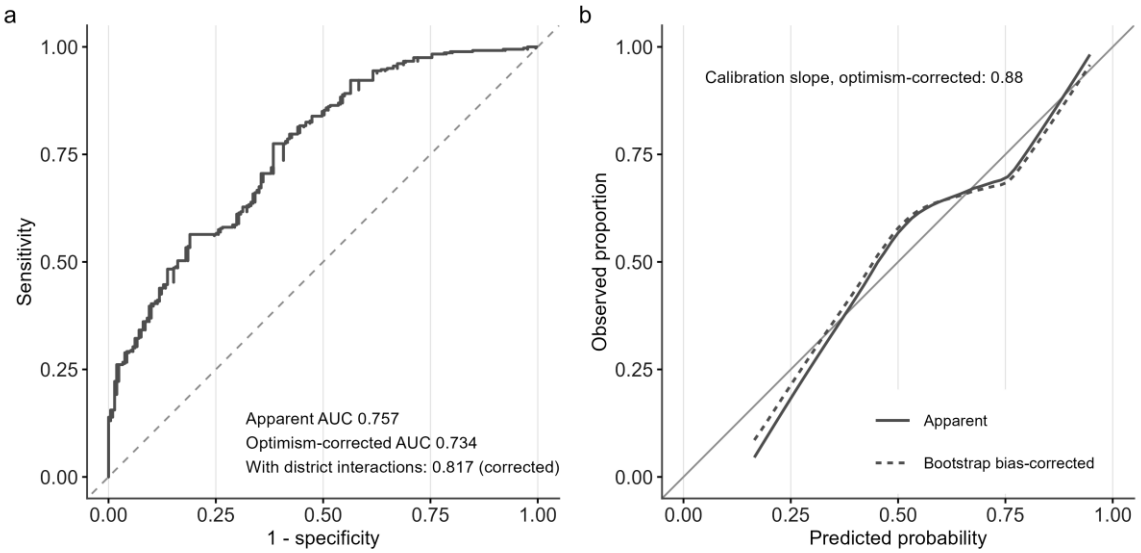


Fig. S1 Discrimination (a) and calibration (b) of the primary model

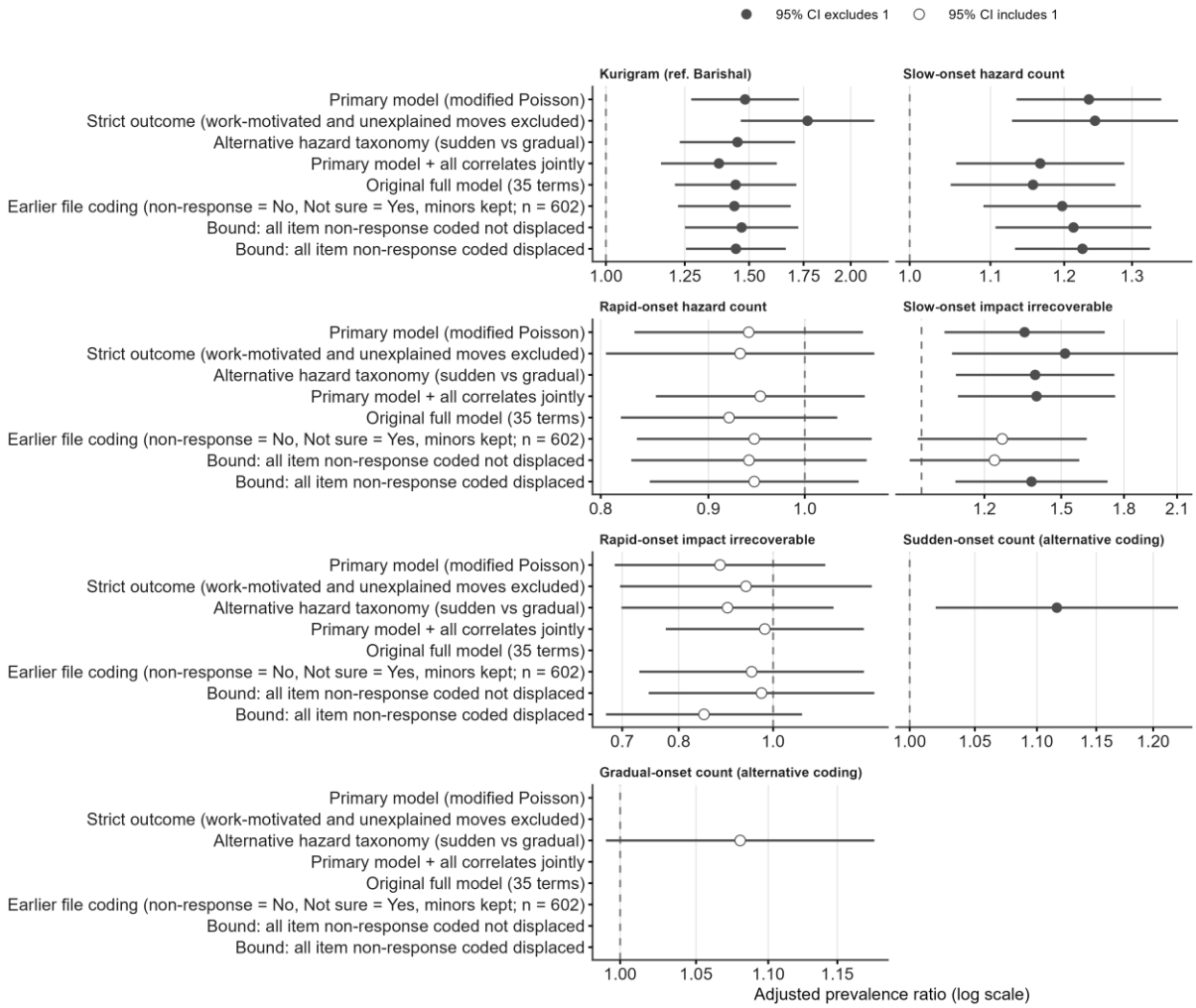


Fig. S2 Pooled adjusted prevalence ratios across specifications and outcome codings

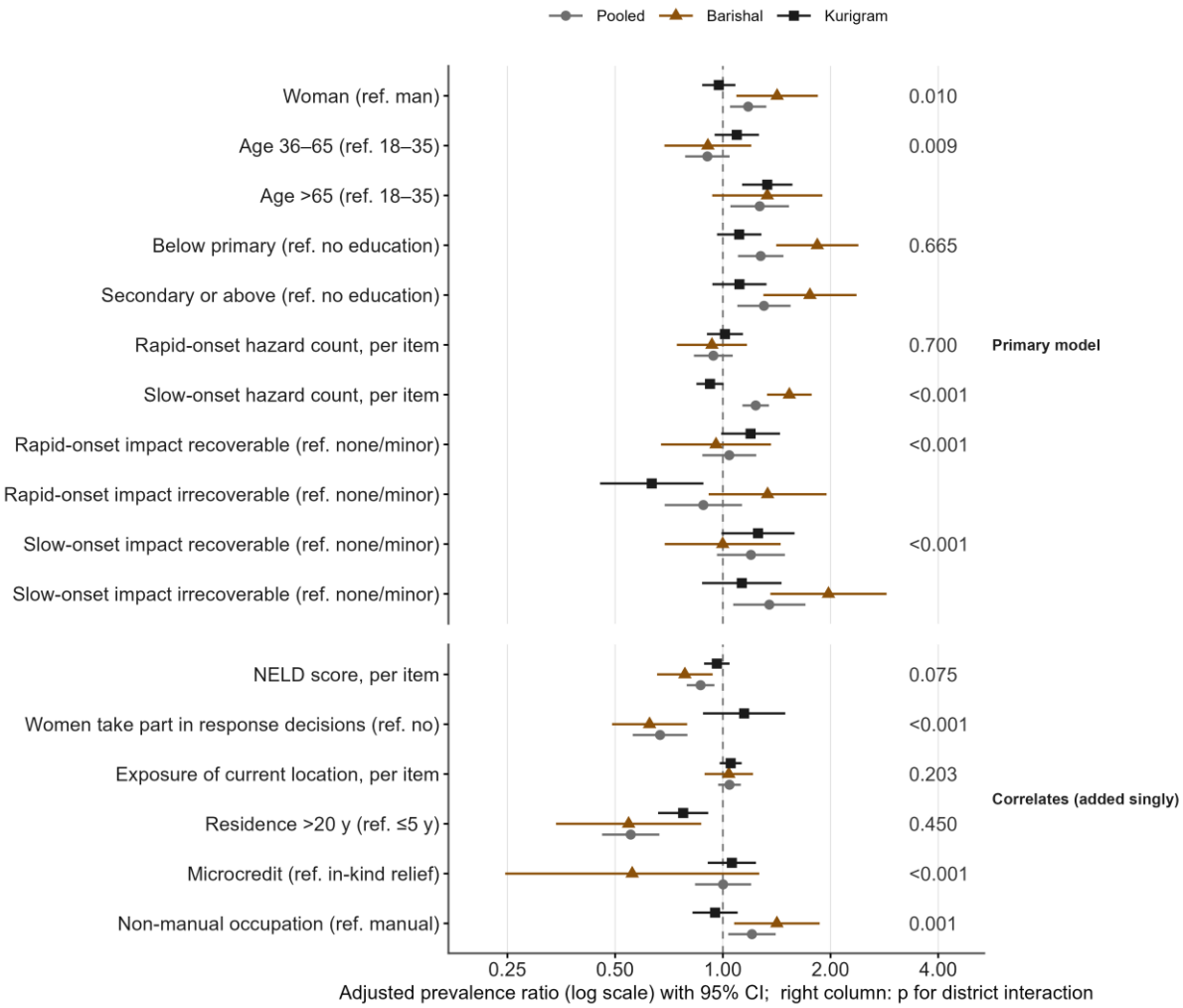


Fig. S3 Adjusted prevalence ratios (95% CI), pooled and by district, for the primary model and for each correlate added singly. Right column: *p* for the district interaction

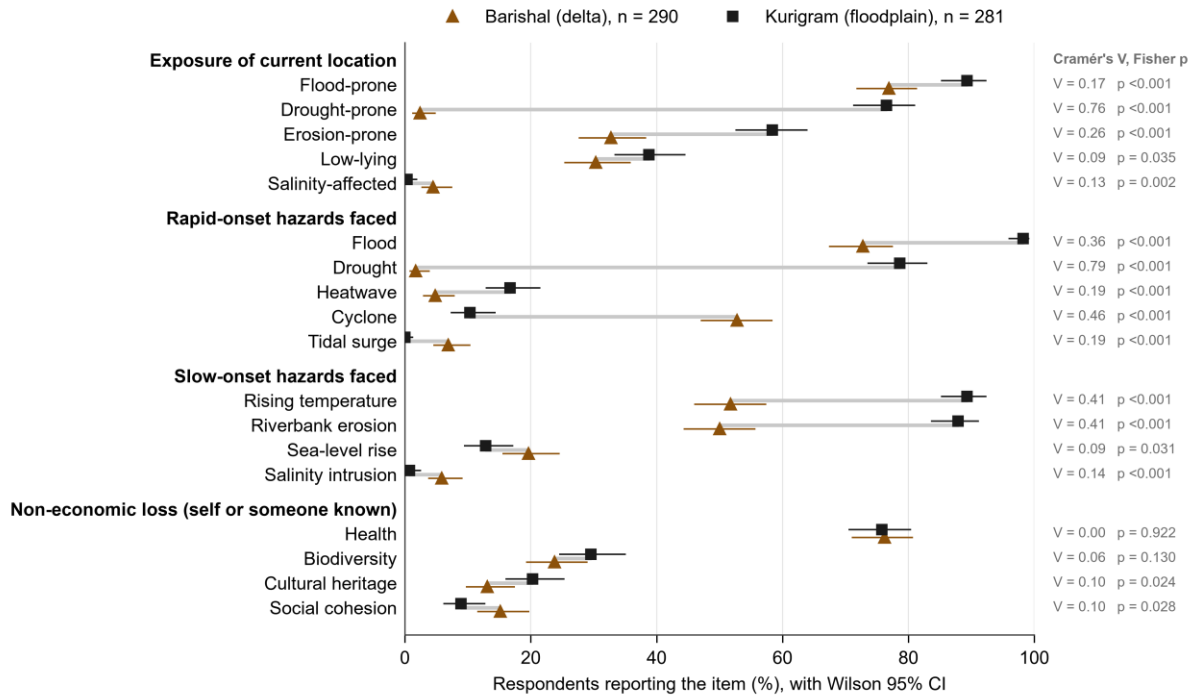


Fig. S4 Exposure, hazard and non-economic loss items by district (Wilson 95% CI)

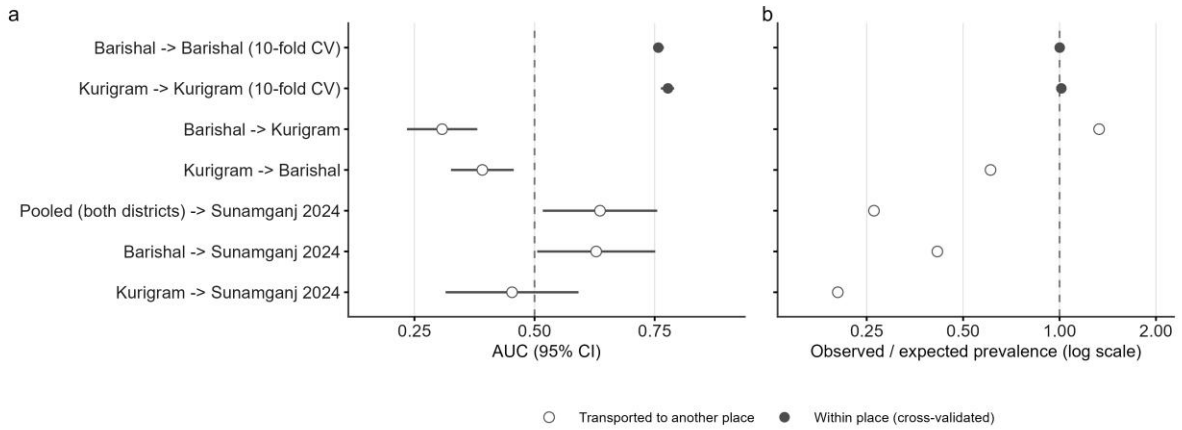


Fig. S5 Transportability of the primary model: AUC (a) and observed over expected prevalence (b)

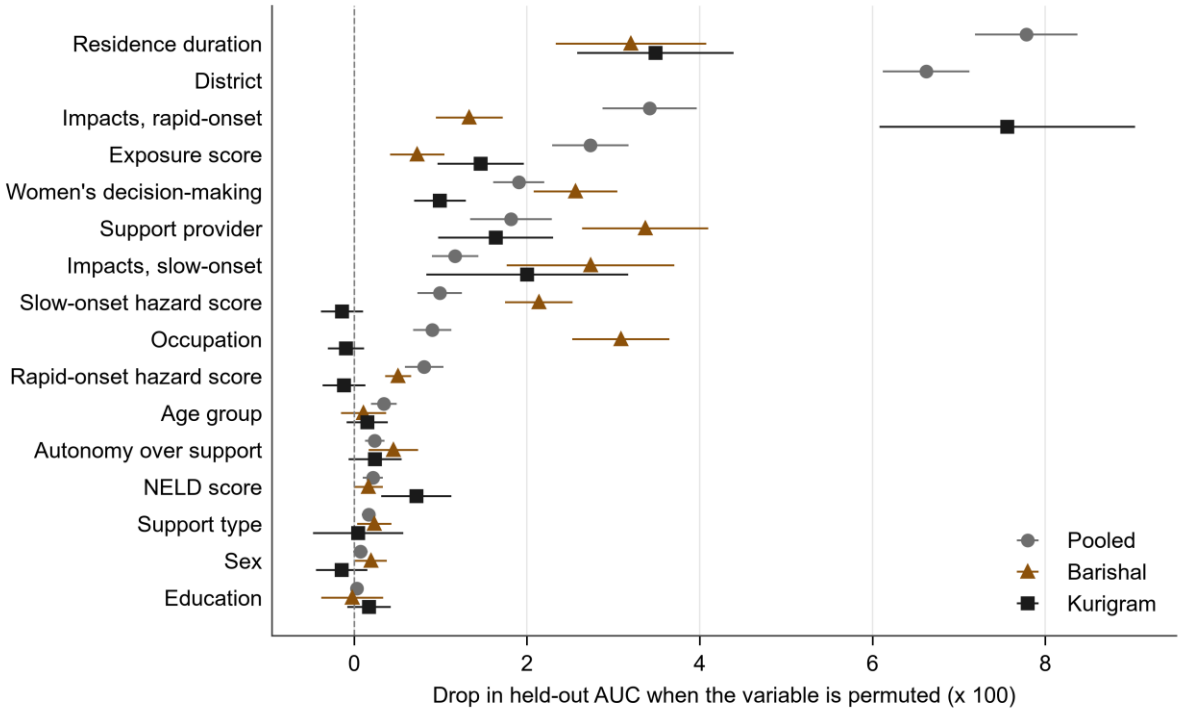


Fig. S6 Grouped permutation importance from random forests (drop in held-out AUC)